UNIVERSITY COMMUNICATIONS ENGINEERING LABORATORY

ANALOG COMMUNICATIONS VISUALIZER VIRTUAL LABORATORY (EC401)

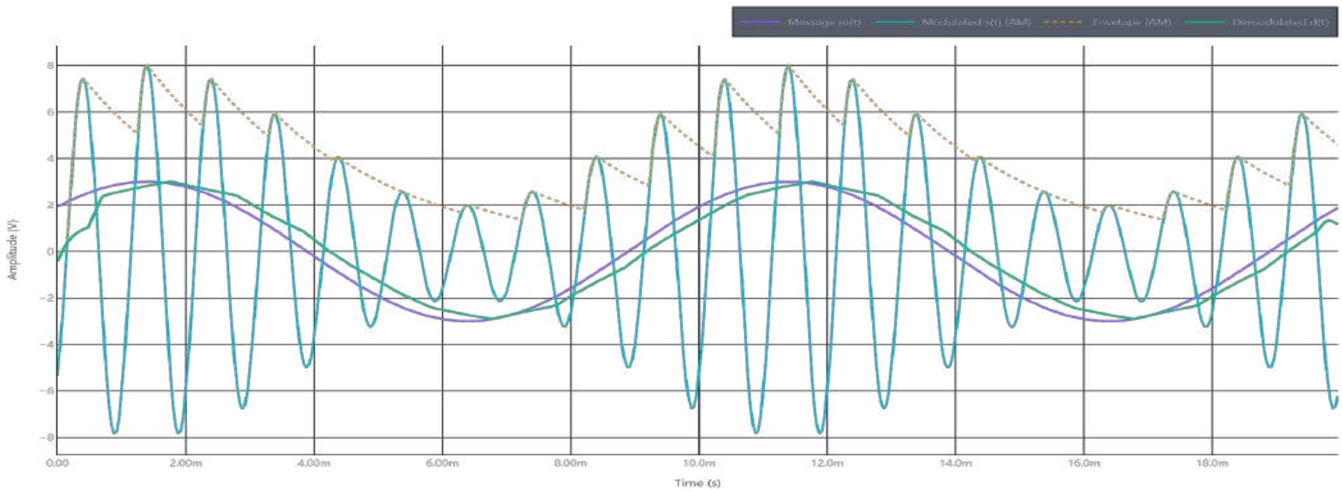
Student Name: Ahan Nath	**Date & Time: 6/9/2026, 11:14:46 PM**
Roll Number: 34900325069	**Modulation: AM (ENVELOPE)**
Subject Code: EC401	

EXPERIMENT: STUDY OF AMPLITUDE MODULATION

1. Initial Configuration Parameters

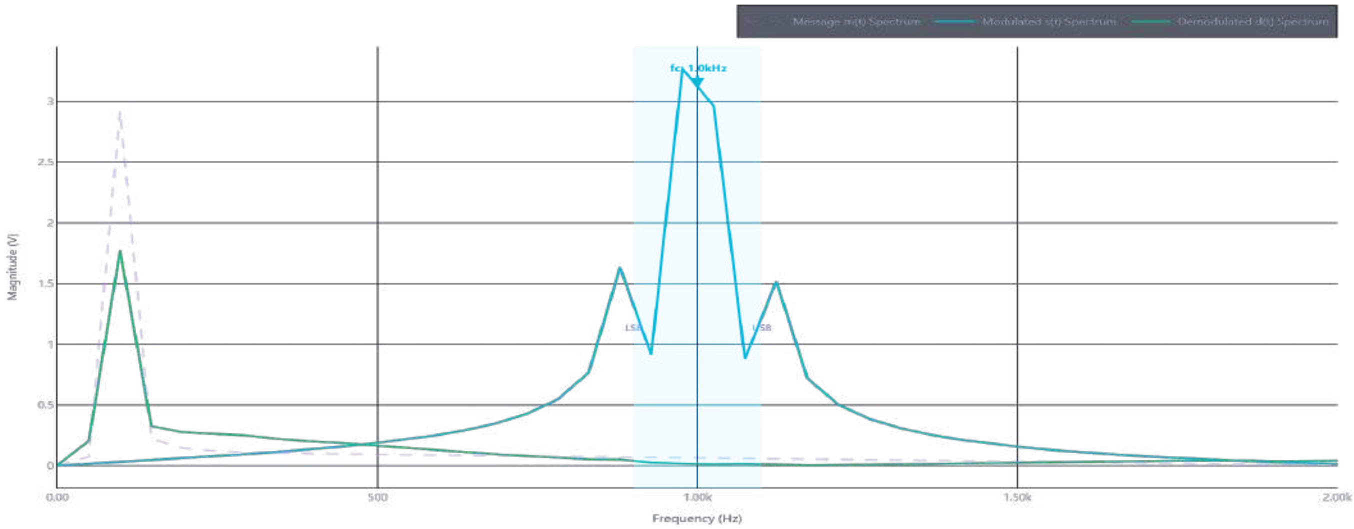
Msg Amplitude (Am)	3.0 V	Carrier Freq (fc)	1.00 kHz
Msg Freq (fm)	100 Hz	Carrier Amplitude (Ac)	5.0 V
Modulation Index (mu)	0.60	Sideband Power	2.250 W
Sampling Freq (fs)	25.00 kHz	Total Power (Pt)	14.750 W
Noise (SNR)	Disabled	Efficiency (eta)	15.3 %

3. Laboratory Oscilloscope Waveform Capture



FREQUENCY SPECTRUM & STUDENT CONCLUSION

4. Spectrum Analyzer FFT Power Spectrum Capture



5. Experiment Conclusion & Diagnostics Summary

We have successfully observed the modulation and demodulation of AM signals. The modulation index was measured to be 0.60, matching the theoretical calculations. The demodulated wave matches the input baseband message with a recovered signal accuracy of 82.4%. Envelope detection suffers diagonal clipping when RC constants are mismatched, whereas coherent synchronous detection yields zero distortion.

LABORATORY EVALUATION & SIGN-OFF

Marks Awarded: ________ / 10 Faculty Name: ________________________ Signature: ________________________

Viva Remarks: ____